Prediction of Bilingual Proficiency through Eye-Tracking Analysis of Mandarin Chinese-English Code-Switching: A Machine Learning Approach

Bibliography

Bosma, E., & Nota, N. (2020). Cognate facilitation in Frisian–Dutch bilingual children's sentence reading: An eye-tracking study. Journal of Experimental Child Psychology, 189, 104699. https://doi.org/10.1016/j.jecp.2019.104699

Buxton, C. (1999). Designing a model-based methodology for science instruction: Lessons from a bilingual classroom. Bilingual Research Journal, 23(2–3), 147–177. https://doi.org/10.1080/15235882.1999.10668684

Chinese as a second language assessment. (2017). Springer Singapore. https://doi.org/10.1007/978-981-10-4089-4

Contemori, C., & Dussias, P. E. (2019). Prediction at the discourse level in Spanish–English bilinguals: An eye-tracking study. Frontiers in Psychology, 10. https://doi.org/10.3389/fpsyg.2019.00956

Cop, U., Drieghe, D., & Duyck, W. (2015). Eye movement patterns in natural reading: A comparison of monolingual and bilingual reading of a novel. PLOS ONE, 10(8), e0134008. https://doi.org/10.1371/journal.pone.0134008

De Cat, C. (2018). Predicting language proficiency in bilingual children. Center for Open Science. https://doi.org/10.31219/osf.io/f5q98

Freeman, M. R., & Marian, V. (2021). VISUAL WORD RECOGNITION IN BILINGUALS. Studies in Second Language Acquisition, 44(3), 759–787. https://doi.org/10.1017/s027226312100053x

Ghasemi, A. A. (2018). Ideal L2 Self, Visual Learning Styles, and L2 Self Confidence in predicting Language Proficiency and L2WTC: A Structural Equation Modeling. English Teaching & Learning, 42(2), 185–205. https://doi.org/10.1007/s42321-018-0010-8

Godfroid, A. (2019). Eye tracking in second language acquisition and bilingualism: A research synthesis and methodological guide. Routledge.

Guerrero, S. L., Whitford, V., Mesite, L., & Luk, G. (2021). Text complexity modulates Cross-Linguistic sentence integration in L2 reading. *Frontiers in Communication*, *6*. https://doi.org/10.3389/fcomm.2021.651769

Kaushanskaya, M., & Marian, V. (2007). Bilingual language processing and interference in bilinguals: Evidence from eye tracking and picture naming. Language Learning, 57(1), 119–163. https://doi.org/10.1111/j.1467-9922.2007.00401.x

Minor, S., Mitrofanova, N., & Westergaard, M. (2024). The interaction of linguistic and visual cues for the processing of case in russian by russian‐german bilinguals: An eye tracking study. Topics in Cognitive Science. https://doi.org/10.1111/tops.12724

Mou, X., He, C., Tan, L., Yu, J., Liang, H., Zhang, J., Tian, Y., Yang, Y.-F., Xu, T., Wang, Q., Cao, M., Chen, Z., Hu, C.-P., Wang, X., Liu, Q., & Wu, H. (2024). ChineseEEG: A Chinese linguistic corpora EEG dataset for semantic alignment and neural decoding. Scientific Data, 11(1). https://doi.org/10.1038/s41597-024-03398-7

Pan, Y. (2014). Adapting English newspapers to develop language proficiency in low-level EFL learners. English Linguistics Research, 3(1). https://doi.org/10.5430/elr.v3n1p66

Pérez, A. I., Schmidt, E., & Tsimpli, I. M. (2023). Inferential evaluation and revision in L1 and L2 text comprehension: An eye movement study. *Bilingualism Language and Cognition*, *27*(4), 585–598. https://doi.org/10.1017/s136672892300069x

Quiñonez-Beltran, J. F., Seymour, T. M., Robbins, R. A. J., Xu, Y., & Joshi, R. M. (2024). What can eye movements tell us about reading in a second language: A scoping review of the literature. Education Sciences, 14(4), 375. https://doi.org/10.3390/educsci14040375

Shalileh, S., Kairov, M., Baminiwatte, R., Parshina, O., & Dragoy, O. (2024). Predicting first-language and second-language proficiency using eye fixation data and demographic information: Assumptions, data representations, and methods. IEEE Access, 12, 145832–145844. https://doi.org/10.1109/access.2024.3468460

Soares, A. P., Oliveira, H., Ferreira, M., Comesaña, M., Macedo, A. F., Ferré, P., Acuña-Fariña, C., Hernández-Cabrera, J., & Fraga, I. (2019). Lexico-syntactic interactions during the processing of temporally ambiguous L2 relative clauses: An eye-tracking study with intermediate and advanced Portuguese-English bilinguals. PLOS ONE, 14(5), e0216779. https://doi.org/10.1371/journal.pone.0216779

Tywoniw, R. (2023). Compensatory effects of individual differences, language proficiency, and reading behavior: An eye-tracking study of second language reading assessment. Frontiers in Communication, 8. https://doi.org/10.3389/fcomm.2023.1176986

Wang, A., & Pellicer‐Sánchez, A. (2022). Incidental vocabulary learning from bilingual subtitled viewing: An eye‐tracking study. Language Learning, 72(3), 765–805. https://doi.org/10.1111/lang.12495

Wang, T., Wang, Y., Hu, H., Wang, X., Chen, S., & Yang, Y. (2025). An eye-movement database of bilingual language control for Chinese-English bilinguals. Scientific Data, 12(1). https://doi.org/10.1038/s41597-025-04628-2

Zemblys, R., Niehorster, D. C., Komogortsev, O., & Holmqvist, K. (2017). Using machine learning to detect events in eye-tracking data. Behavior Research Methods, 50(1), 160–181. https://doi.org/10.3758/s13428-017-0860-3